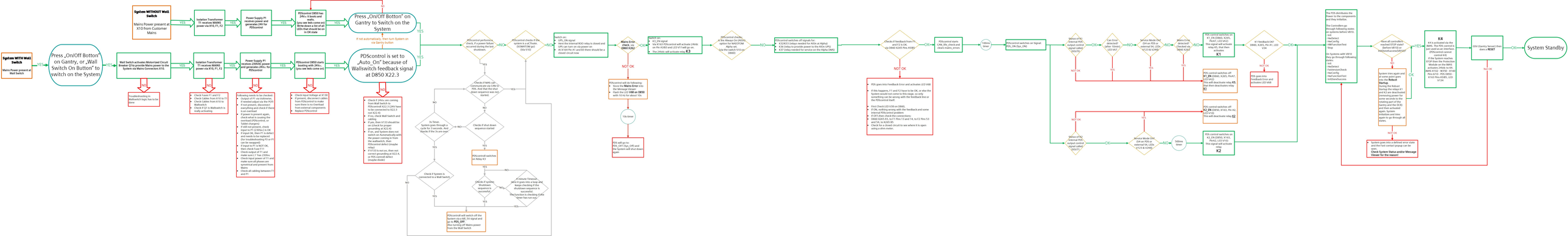
Switch WITH Wall Switch
Mains Power present at Wall Switch
YES
Press „On/Off Button" on Gantry, or „Wall Switch On Button" to switch on the System
YES
Wall Switch activates Motorized Circuit Breaker QT to provide Mains power to the System via Mains Connectors X10.
NO
Troubleshooting in Wallswitch logic has to be done
YES
Isolation Transformer T1 receives MAINS power via X10, F1, F2.
NO
Check Fuses F1 and F2 and replace if necessary to T1
Check Cables from X10 to Wallswitch
Check if Q1 is Wallswitch is really activating
YES
Power Supply P1 receives Mains Voltage and generates 24Vcc for PDScontrol
NO
Following needs to be checked:
Check if P1 is switched on?
If needed adjust via the POT.
If not present, disconnect everything and check if there is an overload.
If overload present again, check what is causing the overload (PDScontrol, or Wallswitch logic).
If still not present, check input to P1 (230Vac) is OK.
If input to P1 is not present and needs to be replaced (for troubleshooting P3 or P1 can be swapped)
If input to P1 is NOT OK, check Mains Input.
Check output of T1 and make sure 1.Y has 230Vac.
Check Mains Input to T1 and make sure all phases are symetrical and present from Mains
Check all cabling between T1 and P1
YES
PDScontrol DB50 has 24Vcc. It boots and waits.
(you see leds come on)
Write down a list of all leds that are present in OK state
YES
Press „On/Off Button" on Gantry to Switch on the System
If not automatically, turn System on via Gantry button
YES
PDSControl is set to „Auto_On" because of Wallswitch feedback signal at D850 X22.3
NO
System WITHOUT Wall Switch
Mains Power present at X10 from Customer Mains
YES
Isolation Transformer T1 receives MAINS power via X10, F1, F2.
NO
Check Fuses F1 and F2 and replace if necessary to T1
Check Cables from X10 to Wallswitch
Check if Q1 is Wallswitch is really activating
YES
Power Supply P1 receives Mains Voltage and generates 24Vcc for PDScontrol
NO
Following needs to be checked:
Check if P1 is switched on?
If needed adjust via the POT.
If not present, disconnect everything and check if there is an overload.
If overload present again, check what is causing the overload (PDScontrol, or Wallswitch logic).
If still not present, check input to P1 (230Vac) is OK.
If input to P1 is not present and needs to be replaced (for troubleshooting P3 or P1 can be swapped)
If input to P1 is NOT OK, check Mains Input.
Check output of T1 and make sure L.Y has 230Vac.
Check Mains Input to T1 and make sure all phases are symetrical and present from Mains
Check all cabling between T1 and P1
YES
PDSControl DB50 fails booting with 24Vcc. (you see leds come on)
YES
Check Input Voltage at X1:30 and Wallswitch feedback signal from PDScontrol to make sure there is no Overload from external components
Replace PDScontrol
NO
Check if 24Vcc are coming from Wall Switch to PDScontrol. Should 24V have to be connected to A22:3 not X22:4!
Check Wall Switch and cabling.
If yes, then V113 should be on (check for proper grounding at X22:4).
If V113 is not on, does not switch on Automatically with the power coming in from Wallswitch (check PDScontrol defect (maybe relay)
If V113 is not on, then not correct grounding at X22:4, or PDScontrol defect (maybe diode)
PDScontrol performs a check, if a power failure occurred (maybe due to shutdown)
YES
Checks if MAN can Communicate via CAN bus (Is MAN ready, or power sequence was not started)
NO
Checks if shut down sequence started
YES
PDScontrol switches on Relay K3
YES
Checks if system shutdown sequence is successful
NO
A minute Timeout. (wait it goes into a loop and keeps checking if the shutdown sequence is successful)
The function is checking if the timer has run out.
YES
PDScontrol will switch off the Mains Power by signal and turn to POS_OFF.
Also turning off Mains power from the Wall Switch
NO
Checks if system is in a Shutdown (Shutdown Input (Xio V10)
YES
3s Timer. System goes through a cycle for 3 seconds. And "Hacks if the 3s are over
NO
Check if System is connected to a Wall Switch
NO
YES
NO
PDSconnect checks if the system is in a Shutdown (Shutdown Input (Xio V10)
NO
Switch on:
UPS_ON signal
Here the internal RO0 relay is closed and the UPS is switched on.
X2:E144 Pin X1 and B1:Bwn should be a closed circuit now
Mains Enter check via D860 X2b2
OK
X13_ON signal
Ha V113 PDScontrol will activate (24Vdc on this A1B0 and LED V1 will go on.
The 24Vdc will activate relay K3
NO
PDSconnect checks, if the Always On (MOD), option for NASDTOM is ON (Xio V9)
Use the switch V10 on D860
NO
PDSconnect switches off signals for:
K32/K33 (relays needed for AEOn at Alpha)
K26 (relay needed for switch on the Alpha PMS)
K27 (relay needed for service on the Alpha DMS)
Checks if feedback from P1 and F2 is OK. Via D860 X20 Pins A5/B5
OK
PDSconnect starts CAN_Sh_check and clears mains_errors
100ms Timer
PDSconnect switches on Signal: POS_ON (Xio_ON)
NOT OK
PDSconnect will do following:
Store the Mains Error into the Error Logbook
Flash the LED V98 on D850 with 10 Hz for about 10s
10s timer
POS will go to: POS_OFF (Xio_OFF) and PDSconnect will shut down again
NOT OK
PDS goes into Feedback Error and activates LED V98
If this happens, F1 and F2 have to be OK, or else the System would not come to this stage, so only something can be wrong with the feedback line or the PDSconnect itself.
First Check LED V30 on D860, if ON, nothing wrong with the feedback and some Internal PDScontrol problem
If V30 is off, check following connections: D860 A20:B5, to F1 Pins 13 and 14, to F2 Pins 53 and 54, to A2:B5 B5
Check for a closed circuit to see where it is open using a ohm meter.
Mains Error detected! (V30 W1)
OK
Can Error detected? (V30 W1)
NO
Service Mode On? (S4 on PDS or remotely via Xio V13 & V34B)
YES
Mains Error detected! (V30 W1) D860 X2b2
YES
PDS control switches off K1_ON (D860, X2b5, Pin A1, LED V42) This will deactivate relay K1 that then deactivates relay K1
PDS control switches off K2_ON (D860, X1b5, Pin A2, LED V10) This will deactivate relay K2
Mains Error detected! (V30 W1)
OK
Service Mode On? (S4 on PDS or remotely via Xio V13 & V34B)
NO
500ms timer
PDS control switches on K2_ON (D850, X1b5, Pin A2, LED V10) This signal will activate relay K2
K1 Feedback OK? (D860, X2b5, Pin A1, LED V34
OK
NOT OK
PDS goes into Feedback Error and activates LED V68
The PDS distributes the Power to the components and they initialize.
The Connection go through following states as systems before VE10, etc:
Init
InitDriver
InitConfig
SWFunctionTest
Stop
Or Systems with VE10 they go through following states:
Init
InitDriver
FanInvertorCheck
InitConfig
HwFunctionTest
InitDoneNetwork
Have all controllers reached STOP state? (before VE10 or InitDoneNetwork)
NO
System tries again and at some point goes into the Reboot Startup
During the Reboot Startup the Relays K1 and K2 are deactivated (removing power for safety reasons) and the rotating part of the Gantry and the DCR2 and there shaped again. System restarts and tries again to go through all states.
NOT OK
System goes into a defined error state and the Fail contact popup can be used.
Check System Status and/or Message Viewer for the reason!
OK
K4
K4 is activated via the IMPS. The PDS control is just used as an interface (PDSconnect cannot switch K4 off)
If the System reaches STOP then the Protection K4 is activated (K4 is activated 24Vdc to K4, MAN: X172 - W170 - X140 Pins 6/10 - PDS D850 - X1b5 Pins A5/B5, LED V114
YES
GEV (Gantry Server) then does a RESET
NO
OK
System Standby